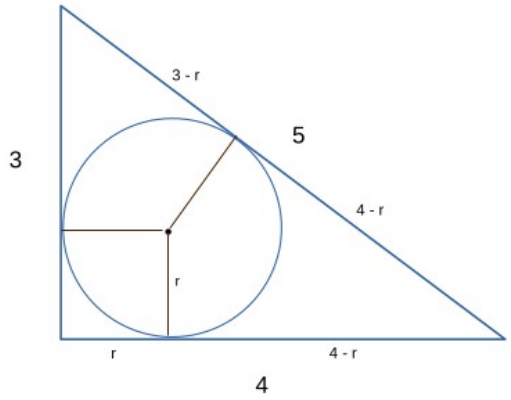
3,4,5 Problem[1]

Determine area of the circle inscribed within a right-angle triangle with sides 3, 4 and 5. What proportion is this area to that of the triangle?

*

Solution

Constructing the figure as described, append radii of the inscribed circle to the tangential points of the three sides. As such, these radii are normal to the sides of the triangle.



We see that this forms a square bound by the centre of the circle and the right-angle of the triangle. This divides the 3-side and 4-side into segments: $r + (3 - r)$, and $r + (4 - r)$, respectively, where r is the radius of the inscribed circle.

Therefore, by Euclid's twin tangent theorem, the 5-side divides into segments $(3 - r)$ and $(4 - r)$. From this we have –

$$5 = (3 - r) + (4 - r)$$

so for the radius…

$$2r = 7 - 5 \qquad \rightarrow \qquad r = 1$$

and area of circle…

$$A_C = \pi$$

Since the triangle's area A_T is equal to 6, the ratio becomes…

$$A_C / A_T = \pi / 6 \quad \approx 52.36\%$$

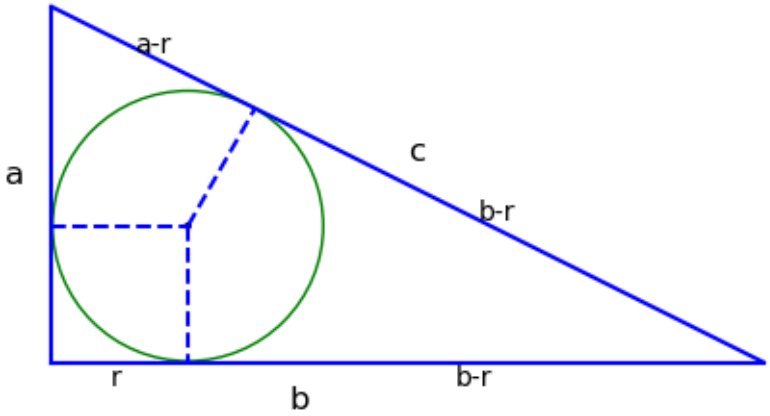
*

Repeating this process with a 5×12×13 right-angle triangle yields radius $r = 2$, with the area ratio = $4\pi / 30$ (approx. 41.89%). A 2×2×√8 example gives $r = 2 - \sqrt{2}$; $A_C = (6 - 4\sqrt{2})\pi$; and ratio approx. 53.90%.

1 From internet website

Generalised Right Triangle

For any right-angle triangle with sides *a*, *b* & *c*, where $a^2 + b^2 = c^2$, we construct…



From which we infer –

$$c = a - r + b - r$$

so…

$$a^2 + b^2 = (a + b - 2r)^2$$
$$= a^2 + 2ab - 4ar - 4br + b^2 + 4r^2$$

thus…

$$0 = 4r^2 - 4(a+b)r + 2ab$$

or…

$$0 = 2r^2 - 2(a+b)r + ab$$

Solving for the quadratic –

$$r = \frac{2(a+b) \pm \sqrt{4(a+b)^2 - 8ab}}{4}$$

$$r = \frac{(a+b) \pm \sqrt{(a^2+b^2+2ab) - 2ab}}{2}$$

$$r = \frac{(a+b) \pm \sqrt{a^2+b^2}}{2}$$

Which is to say…

$$r = ½ (a + b - c)$$

(since the second solution produces a circle larger than the the triangle itself).

*

From this we can determine the circle area A_C –

$$Ac = \pi \frac{(a+b+c)^2}{4} \quad \rightarrow \quad Ac = \pi \frac{(a^2+b^2+ab-(a+b)c)}{2}$$

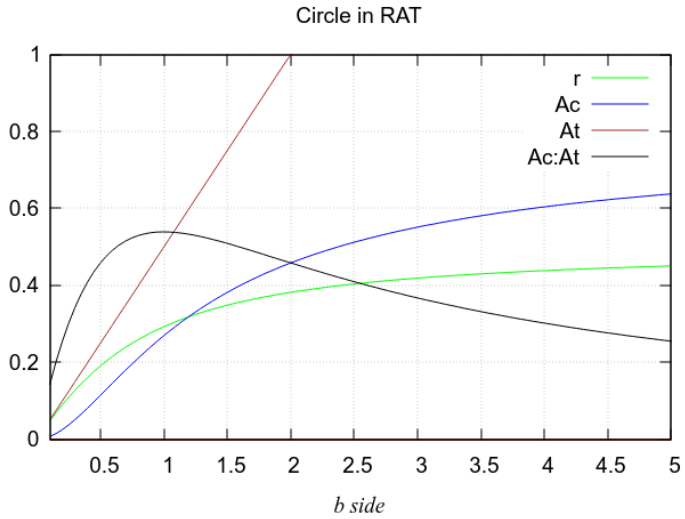
so the ratio…

$$Ac : At = \pi \frac{a^2+b^2+ab-(a+b)c}{ab} \quad = Ar$$

Area Ratio Function

Considering this ratio between the area of of a circle and its circumscribed right angle triangle, we might ask – what is its form, and what shape of triangle yields the greatest proportion for $A_C : A_T$?

Let us scale our figure such that side $a = 1$. Plotting some radius r and area A values for a range of b-side values gives the following graph.



This would suggest that the circle to triangle ratio is maximal where our right triangle is isosceles.

*

To determine the maximal area ratio (Ar), find the derivative of the function above (normalising so that $a = 1$).

$$Ar = \pi \frac{1+b^2+b-(1+b)c}{b} \quad \text{and} \quad c = \sqrt{1+b^2}$$

so...

$$\frac{Ar}{\pi} = \frac{1}{b} + b + 1 - \frac{\sqrt{(1+b^2)}}{b} - \sqrt{(1+b^2)}$$

differentiating[2]...

$$\frac{d}{db} \cdot \frac{Ar}{\pi} = \frac{-1}{b^2} + 1 - \frac{1}{\sqrt{1+b^2}} + \frac{\sqrt{1+b^2}}{b^2} - \frac{b}{\sqrt{1+b^2}}$$

that is…

$$\frac{d}{db} \cdot \frac{Ar}{\pi} = 1 - \frac{1}{b^2} - \frac{\sqrt{1+b^2}}{1+b^2} + \frac{\sqrt{1+b^2}}{b^2} - \frac{b\sqrt{1+b^2}}{1+b^2}$$

so…

$$Ar' = \pi \left(1 + \frac{\sqrt{1+b^2}-1}{b^2} + \frac{b+1}{1+b^2}\right)$$

or...

$$Ar' = \pi \left(1 + \frac{\sqrt{1+b^2}-1}{b^2} - \frac{(b+1)(\sqrt{1+b^2})}{1+b^2}\right)$$

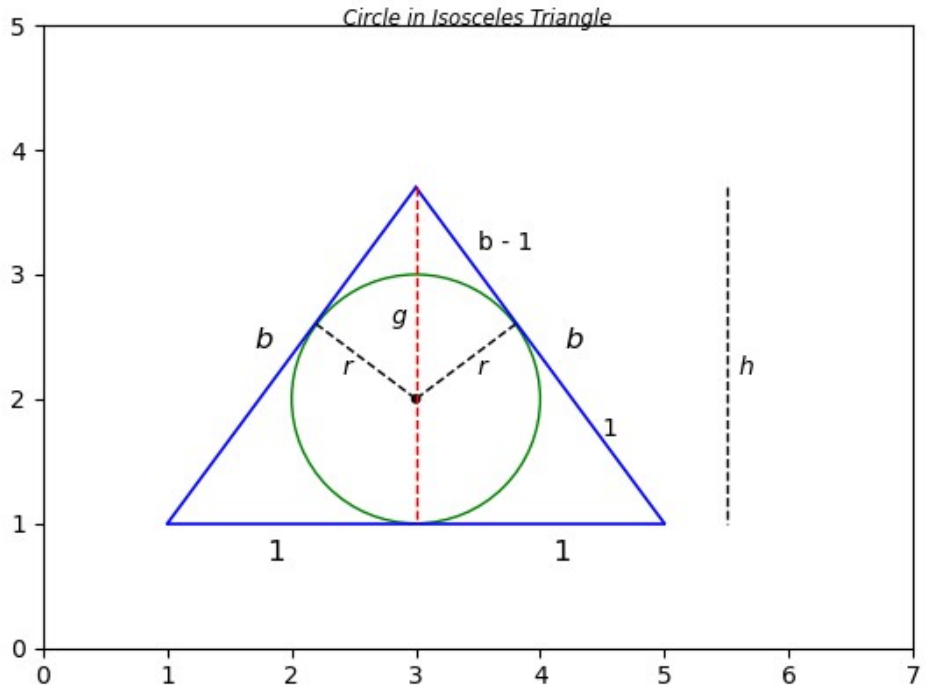
When $b = 1$, the expression inside the brackets becomes – $1 + \frac{\sqrt{2}-1}{1} - \frac{2\sqrt{2}}{2}$ `= 0,` confirming this to be the maximum for the area ratio function.

KDT July 2026

2 By application of chain and division rules for differentiation

Isosceles Triangles

Seeing how the solution above resolves to the right-angle isosceles triangle, let us consider all isosceles triangles in general. With reference to the following diagram, scaling such that the base is 2, we can say $h=\sqrt{b^2-1}$; hence the area of the triangle $A_T=\sqrt{b^2-1}$.



Furthermore –

$$(h-r)^2=r^2+(b-1)^2$$

so…

$$(b^2-1)+r^2-2rh=r^2+b^2+1-2b$$

or…

$$-2rh=2-2b \quad \rightarrow \quad rh=b-1$$

therefore…

$$r=\frac{b-1}{\sqrt{b^2-1}}=\frac{(b-1)\sqrt{b^2-1}}{b^2-1}$$

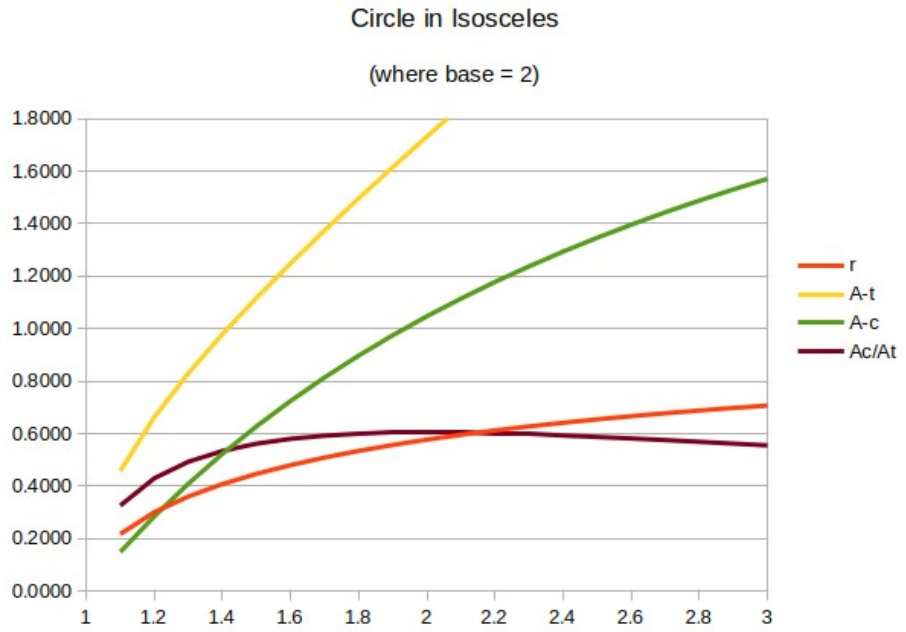
From which we can determine the area of the inscribed circle –

$$A_C=\pi\frac{b^2-2b+1}{b^2-1}$$

and the area ratio –

$$A_R=\pi\frac{b^2-2b+1}{(b^2-1)\sqrt{b^2-1}} \quad \rightarrow \quad \pi\frac{(b^2-2b+1)\sqrt{b^2-1}}{(b^2-1)(b^2-1)}$$

Charting these functions suggests that the maximal area ratio occurs where $b = 2$, when the half-base = 1. Which is to say, the maximum value for A_R is attained when the triangle is equilateral, which seems intuitive.



When $b = 2$, we get circle radius $r = \sqrt{3} / 3$, so its area $A_C = \pi / 3$. Dividing by the triangle's area ($A_T = \sqrt{3}$) yields $\sqrt{3}\,\pi / 9$, approx. 60.46%. (And when b = $\sqrt{2}$ [right triangle] the ratio becomes $\pi(3 - 2\sqrt{2})$, approx. $0.1716\pi \approx 53.90\%$).

*

To differentiate the A_R function above, let us consider the constituant parts. We identify the numerator $numer = (b^2 - 2b + 1) \times \sqrt{b^2 - 1}$; and denominator $denom = (b^4 - 2b^2 + 1)$. Then –

i. derivative of the surd: $\frac{d}{db}\sqrt{b^2-1} = \frac{1}{2}\sqrt{b^2-1}^{(-1/2)} \times 2b \quad \rightarrow \quad \frac{b}{\sqrt{b^2-1}}$ *{chain rule*

ii. remainder of numerator: $\frac{d}{db} b^2 - 2b + 1 = 2b - 2$

iii. deriv. of denominator: $\frac{d}{db} b^4 - 2b^2 + 1 = 4b^3 - 4b$

iv. deriv. of whole of the numerator:

$$\frac{d}{db} numer = \sqrt{b^2-1}(2b-2) + \frac{(b^2-2b+1) \times b}{\sqrt{b^2-1}} \qquad \{product\ rule$$

v. deriv. of whole expression:

$$\frac{d}{db}\frac{numer}{denom} = \frac{denom \times (\sqrt{b^2-1}(2b-2) + \frac{(b^2-2b+1)\times b}{\sqrt{b^2-1}}) - numer \times (4b^3 - 4b)}{denom^2} \qquad \{divn\ rule$$

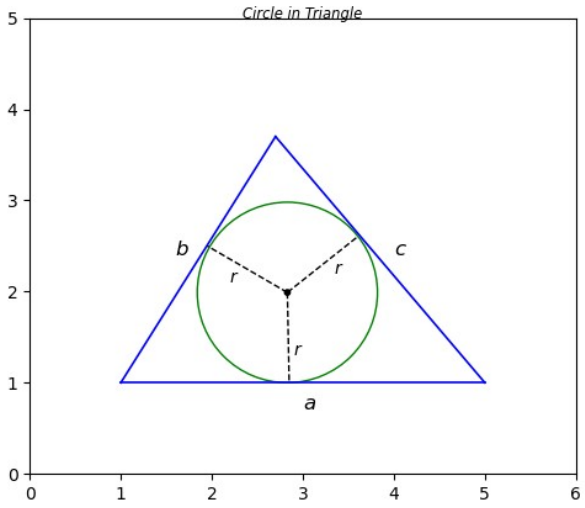
Substituting 2 for b in the top part we get…

$$denom \times (\sqrt{3} \times 2 + \frac{2}{\sqrt{3}}) - numer \times (32 - 8) \quad \rightarrow 9 \times (2\sqrt{3} \times 2 + 2\frac{\sqrt{3}}{3}) - \sqrt{3} \times 24 \quad \rightarrow 24\sqrt{3} - 24\sqrt{3} = 0$$

KDT Aug 2026

All Triangles

Let us finally consider the general case, that is for any scalene triangle. Regards the figure below, a triangle with sides *a*, *b*, *c* enscribes circle radius *r*.



We can divide this figure into three constituant triangles by connecting the apexes to the centre of the circle. Now we see that area of the first of these constituant triangles, based on side *a*, is equal to *ar*/2. Similarly with sides *b* and *c*, so the total area $A_T = ½\ r\ (a + b + c)$.

Now Heron's formula tells us that for the area *A* of a triangle *a*,*b*,*c*…

$$A^2 = s(s-a)\,(s-b)\,(s-c) \text{ where: } \quad s = ½\,(a+b+c)$$

therefore…

$$\frac{r^2}{4} = \frac{½(a+b+c)(s-a)(s-b)(s-c)}{(a+b+c)^2}$$

hence

$$r^2 = \frac{2(s-a)(s-b)(s-c)}{a+b+c}$$

So the area of the circle *squared* becomes…

$$A_c^2 = \frac{4\pi^2(s-a)^2(s-b)^2(s-c)^2}{(a+b+c)^2}$$

and the square of the area ratio of the circle to enclosing triangle reolves to…

$$Ar^2 = \frac{4\pi^2(s-a)^2(s-b)^2(s-c)^2}{(a+b+c)^2 \times s(s-a)(s-b)(s-c)} = \frac{4\pi^2(s-a)(s-b)(s-c)}{(a+b+c)^2 \times s}$$

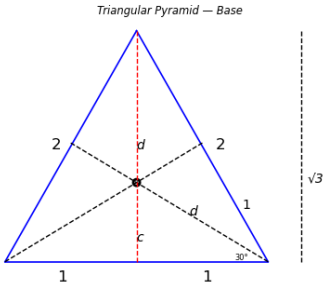
so…

$$Ar = \frac{2\pi\sqrt{(s-a)(s-b)(s-c)}}{\sqrt{s}(a+b+c)} = \frac{\pi\sqrt{s(s-a)(s-b)(s-c)}}{s^2} \text{ or } \frac{4\pi A_T}{(a+b+c)^2}$$

*

In Stereo

We have seen that the most "*efficient*" arrangement of the circle-inscribed triangle, where the area ratio is maximised, is where $a = b = c$, i.e. the equalateral. Out of curiosity, we might ask how does this compare with an equalateral triangular pyramid? Consider the following diagram –



Since $c + d = \sqrt{3}$, and $c = d / 2$, we have $c = \frac{1}{3}\sqrt{3}$ and $d = \frac{2}{3}\sqrt{3}$. So the area of each face is $\sqrt{3}$.
Now the right triangle on d to the apex of the pyramid with height h has an edge with length 2 as its hypotenuse. Hence –

$$h^2 + d^2 = 2^2 \;= 4 - 12/9 = 8/3$$
$$h = \sqrt{8/3} = 2\sqrt{\tfrac{2}{3}} \qquad \textit{(approx. 1.633)}$$

Therefore…

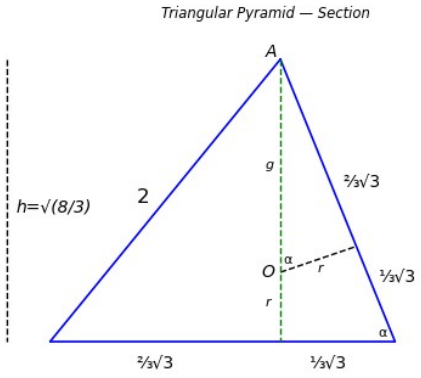
pyramid surface area: $A_P = 4\sqrt{3}$ *(approx. 6.928)*

and

pyr volume: $V_P = \frac{1}{3}\sqrt{3} \times (\sqrt{8}/\sqrt{3}) = \frac{2}{3}\sqrt{2}$ *(approx. 0.943)*

*

Next, take a section through the pyramid such as the base extends from a corner to the centre of the opposite side (as in the dotted lines in the figure above), and the top being the apex of the pyramid. As per the diagram below –



Since the inscribed sphere touches each face at its centre, radii therefrom normal to the faces intersect at O, the sphere's centre. And since this produces two similar right-angle triangles, we can say –

$$g : \frac{2}{3}\sqrt{(3)} = \sqrt{3} : h \;\rightarrow\; g = \frac{2}{h} = \frac{2\sqrt{3}}{\sqrt{8}} = \frac{\sqrt{3}}{\sqrt{2}} \qquad \textit{(approx. 1.225)}$$

so…

$$r = \frac{\sqrt{8}}{\sqrt{3}} - \frac{\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{16} - \sqrt{9}}{\sqrt{6}} = \frac{1}{\sqrt{6}} \qquad \textit{(approx. 0.408)}$$

Which gives surface area and volume for the sphere –

$$A_S = 4\pi \times \frac{1}{6} = \frac{2}{3}\pi \qquad \textit{(~2.094)} \qquad\qquad V_S = \frac{4}{3}\pi \times \frac{1}{6\sqrt{6}} = \frac{2}{9\sqrt{6}}\pi = \frac{\sqrt{6}\pi}{27} \qquad \textit{(~0.285)}$$

and sphere:pyramid ratios –

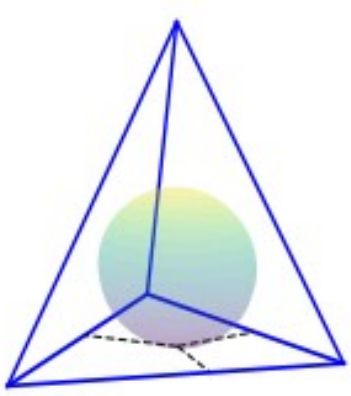
$$A_S / A_P = \frac{2\pi}{3} / 4\sqrt{3} = \frac{\sqrt{3}\pi}{18} \qquad \textit{(~ 0.3023)}$$

$$V_S / V_P = \frac{\sqrt{6}\pi}{27} / \frac{2\sqrt{2}}{3} = \frac{\sqrt{3}\pi}{18} \qquad \textit{(~ ditto)}$$

*

Interesting. By comparison, the corresponding calculations for the sphere inscribed within a cube give –

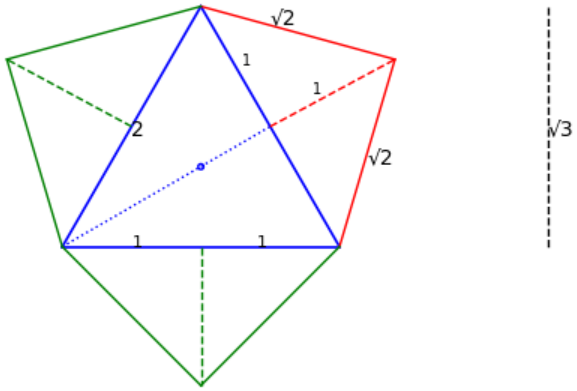
$$A_S / A_C = V_S / V_C = \frac{\pi}{6}$$



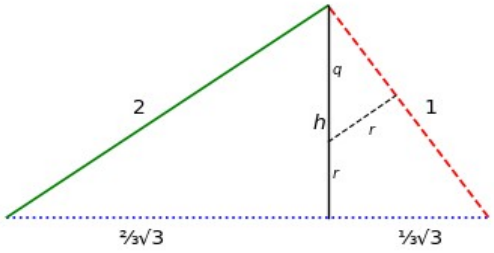
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Right Pyramid

Alternatively, take the pyramid with an equilateral base and the rising edges meeting at right angles. Scaling the base sides to 2, we can surmise that the other edges must equal √2 so as to form right triangles. Unfolding the sides of the pyramid produces the following digram –



Folding up the sides and taking a section through the solid figure, from the left corner, through the apex to the centre of the right side (along the dotted line), results as follows –



From which we determine the height of the pyramid *h*…

$$h^2 = 1 - \tfrac{1}{3} \quad \rightarrow \quad h = \sqrt{(\tfrac{2}{3})} = \tfrac{1}{3}\sqrt{6} \qquad (\sim 0.8165)$$

Now the radius *r* of the inscribed sphere must be such that it meets the rising face (red line) orthagonally, and intersects with the radius from the central point of the base triangle. By dint of the similar triangles in the figure above, we can say that –

$$r / q = \sqrt{3} / 3$$

and since $r = h - q$…

$$\frac{\sqrt{6}}{3} - q = \frac{\sqrt{3}}{3} q \quad \rightarrow \quad \sqrt{6} = q\left(3 + \sqrt{3}\right) \quad \rightarrow \quad q = \frac{\sqrt{6}}{3+\sqrt{3}} = \frac{(3-\sqrt{3})\sqrt{6}}{6} \qquad (\sim 0.5176)$$

therefore…

$$r = \frac{2\sqrt{6} - 3\sqrt{6} + \sqrt{18}}{6} = \frac{3\sqrt{2} - \sqrt{6}}{6} \qquad (\sim 0.2989)$$

*

Volumes for the pyramid and inscribed sphere, and their ratio, can now be calculated –

$$V_S=\frac{4}{3}\pi\frac{(3\sqrt{2}-\sqrt{6})^3}{6^3}=\frac{4}{3}\pi\frac{(18+6-6\sqrt{12})}{36}\frac{(3\sqrt{2}-\sqrt{6})}{6}$$
$$\rightarrow=\frac{1}{3}\pi\frac{(24-12\sqrt{3})}{9}\frac{(3\sqrt{2}-\sqrt{6})}{6}=\frac{\pi}{27}(4-2\sqrt{3})(3\sqrt{2}-\sqrt{6})=\frac{2\pi}{27}(9\sqrt{2}-5\sqrt{6}) \quad \textit{(~ 0.1118)}$$

$$V_P=⅓\sqrt{3}\times\frac{\sqrt{6}}{3}=⅓\sqrt{2} \quad \textit{(~ 0.4714)}$$

$$V_R=V_S/V_P=\frac{2\pi}{27}(9\sqrt{2}-5\sqrt{6})\times\frac{3}{\sqrt{2}}=\frac{2\pi}{9}(9-5\sqrt{3}) \quad \textit{(~ 0.2372)}$$

Likewise their respective surface areas –

$$A_S=4\pi\frac{(3\sqrt{2}-\sqrt{6})^2}{36}=\frac{4\pi}{6}(18+6-6\sqrt{6})=\frac{2}{3}(4-2\sqrt{3})\pi \quad \textit{(~ 1.1224)}$$

$$A_P=\sqrt{3}+3\times1 \quad \textit{(~ 4.7321)}$$

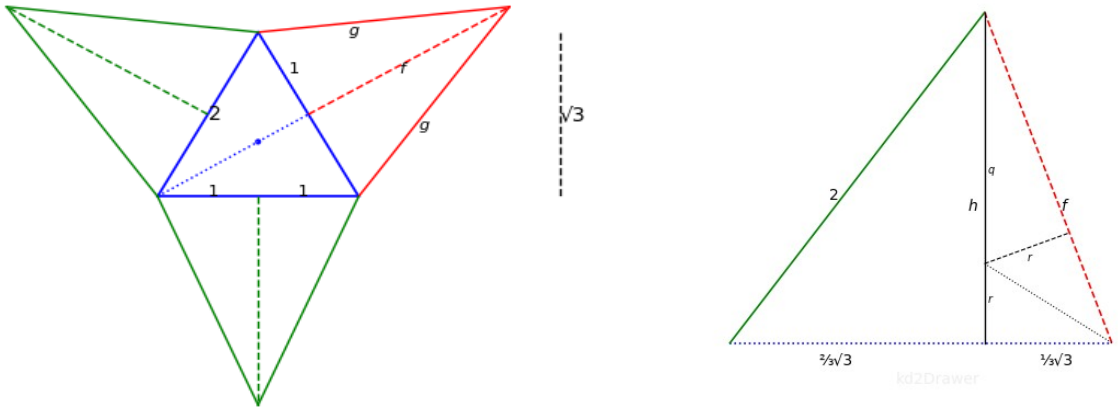
$$A_R=\frac{⅔(4-2\sqrt{3})\pi}{3+\sqrt{3}}=\frac{2(4-2\sqrt{3})\pi}{3}\times\frac{3-\sqrt{3}}{6}$$
$$\rightarrow=\pi(4-2\sqrt{3})(3-\sqrt{3})/9=\frac{2\pi}{9}(9-5\sqrt{3}) \quad \textit{(~ ditto)}$$

Hmm.

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Other Constructs

Next consider the case of the regular tetrahedron previously examined, then drawing up (or pressing down) the top apex point vertically, producing a figure of any chosen height *h*, we form a pyramid comprising an equilateral base and three congruant isoseles triangles on the sides.



Now we can say… $f=\sqrt{h^2+⅓}$ where *f* is the height of the isosoles face triangle (whose base is 2 in our construct). From which we get volume and surface area of this figure –

$$V_P=⅓\sqrt{3}h$$

$$A_P=\sqrt{3}+3f=3\sqrt{h^2+⅓}+\sqrt{3}$$

From the right hand diagram, representing a section through the pyramid, we see the radii *r* of an inscribed sphere, tangential to the blue base and the red face, form a kite-shaped quadralateral. This divides into two right-angle triangles, and since they share a hypotenuse and have another equal side (*r*), their third sides must be equal (i.e. ⅓√3). Therefore the dotted red line *f* divides into a lower part ⅓√3 and the remainder being $\sqrt{h^2+⅓}-⅓\sqrt{3}$.

Whereupon from the similar right triangles within the figure we can determine the spherical radius –

$$\frac{r}{\sqrt{h^2+⅓}-⅓\sqrt{3}}=\frac{⅓\sqrt{3}}{h} \quad \rightarrow \quad r=\frac{⅓\sqrt{3}\sqrt{h^2+⅓}-⅓}{h}$$

which allows for its volume and surface area to be calculated – letting $H = h^2 + ⅓$…

$$V_S=\frac{4\pi}{3}\times\frac{\left(⅓\sqrt{3H}-⅓\right)^3}{h^3} = \frac{4\pi}{3}\times\frac{\left(⅓H-2/9\sqrt{3H}+1/9\right)\left(⅓\sqrt{3H}-⅓\right)}{h^3}$$

$$= \frac{4\pi}{3}\times\frac{\sqrt{3H^3}-3H+\sqrt{3H}-⅓}{9h^3}$$

$$A_S=\frac{4\pi}{3}\times\frac{H-⅔\sqrt{3H}+⅓}{h^2}$$

*

Ratios

We have seen that for the equalateral and right-angled triangular pyramids construed above that the ratio of the surface areas equals that of the volumes, between the pyramid and its inscribed sphere. Which leads to the *conjecture* –

$$\frac{V_S}{V_P}=\frac{A_S}{A_P}$$ where V_X and A_X are the volumes and areas of the respective sphere and pyramid.

If this is to hold for the constructs above (i.e. three isoceles triangles supported by a equilateral base), then it needs be shown that: $V_S \times A_P = V_P \times A_S$. Using the formulation derived above…

[1] $$V_S \times A_P = \frac{4\pi}{3} r^3 \times \left(3\sqrt{H}+\sqrt{3}\right) = \frac{4\pi r^2}{3}\left(3\sqrt{H}+\sqrt{3}\right)\frac{\left(⅓\sqrt{3H}-⅓\right)}{h}$$ *(where $H=h^2$-⅓)*

and

[2] $$A_S \times V_P = 4\pi r^2 \times ⅓\sqrt{3}h = \frac{4\pi r^2}{3}\sqrt{3}h$$

Now products [1] and [2] are equal if and only if…

$$\left(3\sqrt{H}+\sqrt{3}\right)\left(⅓\sqrt{3H}-⅓\right)=\sqrt{3}h^2$$

$$\sqrt{3}H-\sqrt{H}+\sqrt{H}-⅓\sqrt{3}=\sqrt{3}\left(H-⅓\right)$$ *(since $h=\sqrt{(H-⅓)}$)*

Therefore, the relationship posited above holds for all such pyramids of the form described.

KDT Oct 2026